

AARON JONES

Hardware & Embedded Systems Engineering

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PROFESSIONAL SUMMARY

Hardware and embedded-systems professional with experience assembling, bringing up, debugging, and validating electronic systems across wireless power, power electronics, microcontrollers, sensors, and networked devices. Skilled in PCB rework, component-level troubleshooting, oscilloscope/VNA measurements, STM32/ESP32/Raspberry Pi development, and translating electrical findings into verified hardware corrections.

CORE TECHNICAL SKILLS

Hardware: PCB assembly, SMT, soldering, hot-air rework, reflow, board bring-up, analog/digital electronics, power electronics, wireless power, resonant circuits, GaN systems, sensors, motors

Test Equipment: Oscilloscope, VNA, DMM, bench power supply, microscope, continuity/resistance checks, voltage rails, waveform and impedance analysis

Embedded: STM32/NUCLEO, ESP32/ESP32-CAM, Arduino, Raspberry Pi, C/C++, Python, GPIO, I2C, UART, SPI, PWM, interrupts, Wi-Fi, BLE

Development Tools: Git/GitHub, VS Code, PlatformIO, STM32CubeIDE, Linux, MATLAB, Simulink, LabVIEW, NI-VISA, SCPI

ENGINEERING EXPERIENCE

Yank Technologies, Inc.

Electronics Technician - Manufacturing, Test & Engineering Support

Brooklyn, NY

Feb 2026-Present

- Support design, development, debugging, manufacturing, and verification of wireless power and communication hardware, firmware, and system interfaces.
- Assemble and rework PCBs using SMT placement, stencil printing, reflow, soldering, hot-air tools, microscope inspection, and component replacement.
- Perform board bring-up and component-level failure analysis with schematics, DMMs, bench supplies, oscilloscopes, VNAs, rail measurements, waveform inspection, and circuit isolation.
- Troubleshoot resonant power systems, custom inductors/coils, GaN switching stages, reflected impedance, resonant-frequency behavior, and capacitor matching networks.
- Create rework, inspection, pre-power, functional-test, and verification documentation and guide technicians through hardware assembly and troubleshooting.

New York City College of Technology, CUNY

College Lab Technician - Computer Engineering Technology

Brooklyn, NY

June 2023-Present

- Maintain and troubleshoot Arduino, ESP32, STM32/NUCLEO, sensors, power supplies, and embedded lab equipment; diagnose wiring, power, peripheral, and firmware-upload faults.
- Prepare instructional labs and guide students in oscilloscopes, DMMs, GPIO, communication protocols, and embedded development tools.

SELECTED ENGINEERING PROJECTS

Multi-Camera CCTV & Edge Monitoring | Raspberry Pi, ESP32-CAM, Linux

- Built networked camera nodes with centralized viewing, logging, and remote-monitoring workflows; debugged Wi-Fi, streaming, frame-rate, and brownout conditions.

Embedded Control & Sensor Systems | ESP32, Arduino, STM32

- Developed proximity-alert, environmental-monitoring, camera, and RC-control prototypes using sensors, OLEDs, motors, PWM, wireless alerts, and battery monitoring; resolved I2C, grounding, timing, and power-delivery faults.

EDUCATION

New York City College of Technology, CUNY

Bachelor of Technology, Computer Engineering Technology - Magna Cum Laude

Associate in Applied Science, Electromechanical Engineering Technology

Brooklyn, NY

May 2025

June 2023